

NIELIT Virtual Academy

National Institute of Electronics and Information Technology, Chennai

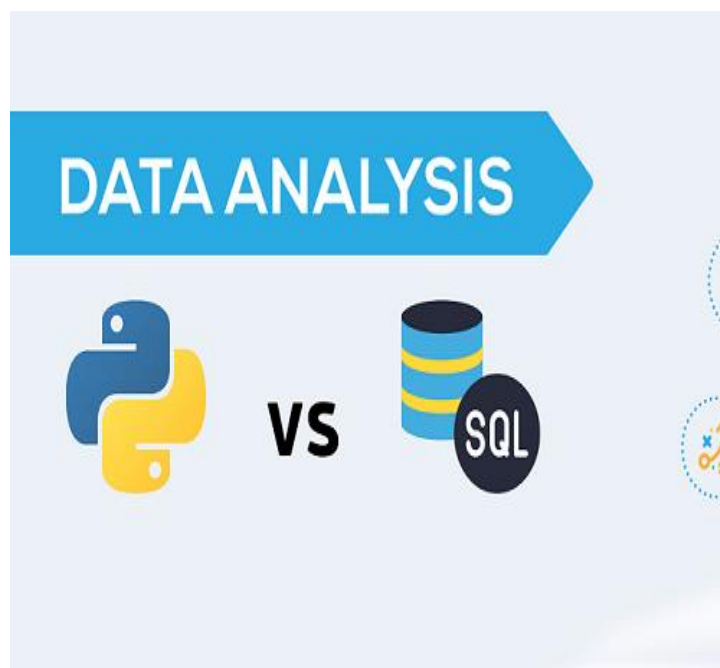
Autonomous Scientific Society of Ministry of Electronics & Information Technology (MeitY), Govt.
of India, ISTE Complex, 25, Gandhi Mandapam Road, Chennai – 600025

Course Prospectus

NSQF Aligned

Mode: ONLINE (Blended)

Data Analysis with Python and SQL



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Course Prospectus

Course Name : Data Analysis with Python and SQL

NSQF Level: 05

Duration: 120 Hours

Last Date of Registration: 05-06-2025

Date of publishing Provisional Selection List: 06-06-2025

Course Start Date: 09-06-2025

Preamble:

The "Data Analysis with Python and SQL" course is designed to provide students with the essential skills and knowledge required to effectively analyze and interpret data using two powerful tools: Python and SQL. This comprehensive course will cover fundamental concepts, practical techniques, and real-world applications, enabling students to handle data from diverse sources, perform data cleaning, manipulation, and visualization, and derive actionable insights. The course is suitable for beginners with no prior experience in data analysis, as well as those looking to enhance their existing skills.

Objective of the Course:

The course objectives for "Data Analysis with Python and SQL" are to provide students with a comprehensive understanding of data analysis, emphasizing its importance and process. Students will master Python basics and essential data analysis libraries, enabling them to perform data cleaning, manipulation, and transformation using Pandas. They will learn to create and customize effective data visualizations. The course also covers SQL fundamentals, teaching students to write queries for data retrieval, filtering, and sorting, and advances to more complex SQL functions and query optimization. Integrating Python and SQL, students will enhance their data analysis capabilities by combining both tools.

Outcome of the Course:

- ✓ Utilize Python and its libraries such as NumPy, Pandas, Matplotlib, and Seaborn to manipulate, analyze, and visualize data.
- ✓ Retrieve, filter, sort, and aggregate data using SQL queries, and employ advanced SQL techniques to perform complex data manipulations and calculations.
- ✓ Design and customize various types of plots to clearly communicate data insights using Matplotlib and Seaborn.
- ✓ Compile and present findings and insights through comprehensive reports and visualizations, effectively communicating the results of their analysis.

Course Structure

S.No	Module Name	Th.	Pr.	Total
1.	Data Modelling Concepts	12.5	12.5	25
2.	DBMS and RDBMS	5	5	10
3.	RDBMS Functionality	5	5	10
4.	NoSQL	7.5	7.5	15
5.	Introduction to Python	7.5	7.5	15
6.	Functions	5	5	10
7.	Classes And Objects	10	10	20
8.	Data Frame Basic Functions	7.5	7.5	15
	Duration (in Hours)	60	60	120

Course Fee Structure:

	General/OBC/EWS	SC/ST	Last Date of Registration
Course Fee	Rs.6000/-	NIL	05-06-2025

Eligibility

S. No.	Academic/Skill Qualification (with Specialization - if applicable)	Required Experience (with Specialization - if applicable)
1	➤ Pursuing Final Year B.Tech in any branch of Engineering , Or ➤ Pursuing Final Year MCA ,Or ➤ Pursuing Final Year B.Sc. in any branch of Sciences , Or ➤ Pursuing Final Year B.Sc. in IT/CS/Electronics/allied subjects Note : 1)Students should have relevant knowledge of the Data Analysis and programming concepts. 2)Students with the above entry requirements are eligible to take the course subject to clearing the written test comprising of Analytical Reasoning, Mathematics and English 3)Passout students in the above entry requirements are also eligible for the course.	-NA-

Number of Seats: 80(Eighty) – Total

Note: Seats are allocated based on the merit of the Qualification

How to Apply?

Candidates can apply online in our **website <http://nva.nielit.gov.in>**. Payment towards non-refundable registration fee can be paid through any of the following modes:

- ✓ **Payment Gateway**
- ✓ **Online transaction: Account No: 31185720641 Branch: Kottur (Chennai), IFSCode: SBIN0001669.**
- ✓ **GPAY/any UPI, Credit Card**

Note: The Institute will not be responsible for any mistakes done by either the bank concerned or by the depositor while remitting the amount into our account

Last date of Registration: 05-06-2025

Selection Criteria :

Selection of candidates will be based on their marks in the qualifying examination subject to eligibility and availability of seats.

- ✓ The first list of Provisionally Selected Candidates will be published on NIELIT Chennai website (www.nielit.gov.in/chennai/index.php) on **06-06-2025 by 5:00 PM**. In case of vacancy, an additional selection list will be prepared and the selection will be intimated by email only.
- ✓ Following documents of candidates will be verified:
 - Qualifying Degree (Consolidated Marksheet/Degree Certificate/Course Completion Certificate), 10th and 12th mark sheet.
 - One passport size photograph.
 - Self-attested copy of Govt. issued photo ID card.
 - AADHAR Copy
- ✓ Selected candidates are requested to upload the proof of remittance of fee on registration portal and also send the proof of remittance of fee as email to **trng.chennai@nielit.gov.in/dhirendra@nielit.gov.in**

Admission:

All provisionally selected who have paid the fees and verified by accounts section of NIELIT Chennai will get a welcome message in his login id provided during registration.

Note-All Provisionally Selected Candidates have to visit NIELIT Chennai for Certificate Verification.

Otherwise their candidature will be cancelled without any intimation.

The credentials and URL for online portal will be shared through WhatsApp or email.

Discontinuing the course

- ✓ No fees under any circumstances, shall be refunded in the event of a student who have completed the process of admission or discontinuing the course in between. No certificate shall be issued for the classes attended.
- ✓ If candidates are not uploading consecutive 3 assignments within assigned time, then their candidature will be cancelled without any notice and all fees paid will be forfeited.
- ✓ If candidates are not appearing for any internal examinations/practical their candidature will be cancelled without any notice and all fees paid will be forfeited.

Course Timings:

This program is a practical oriented one and hence there shall be more lab than theory classes. The classes and labs are online cloud-based from **02 pm to 05 pm** and Monday to Friday.

Course enquiries

Students can enquire about the various courses either on telephone or by personal contact between 9.15 A.M. to 5.15 P.M. (Lunch time 1.00 pm to 1.30 pm) Monday to Friday.

Placement:

Students who have completed the course successfully and qualified, Placement guidance and career counselling will be given to assist in their interviews.

Important Dates

Last Date of Registration: 05-06-2025

Display of Provisional Selection List: 06-06-2025

Course Start Date: 09-06-2025

Examination & Certification

- ✓ Final Certificates will be issued after successful completion of all the modules including mini project. For getting certificate a candidate has to pass each module individually with minimum required marks of 50%.

NSQF Examination Pattern:

➤ Means of assessment:

S. No	Examination Pattern	Modules Covered	Duration in Minutes	Maximum Marks
1.	Theory 1: Data Analysis with Python and SQL	All	90	100
4	Practical 1: Data Analysis with Python and SQL	All	120	60
6	Internal Assessment		-	20
7	Project/Presentation /Assignment		-	20
Total				200

Examination Centre: NIELIT Chennai, Mode: Online

Grading Scheme

Following Grading scheme (on the basis of total marks) will be followed:

Grade	S	A	B	C	D
Marks Range (in %)	$\geq 85\%$	$\geq 75\%$ and $< 85\%$	$\geq 65\%$ and $< 75\%$	$\geq 55\%$ and $< 65\%$	$\geq 50\%$ and $< 55\%$

Testimonials:

“

EXCEPTIONAL SWITCH FOR MY PROFFESION.

Presently I am working as Big Data Engineer at Cognizant, a US based company. The PG Diploma in Data Science and Analytics course proved to be an added advantage in changing my career from SAP to a promissing Data Engineering field.

-SUMIRAN KUMAR SINHA
Big Data Engineer



“

PG program in Data Science and Analytics had excellent content and assignments that build on your knowledge, reinforce, and then expand, the mentoring approach was good, and working with other people via the Internet was good. The instructors did a good job of communicating and making it a more intimate arrangement. A lot of online courses fail because of the isolation. PGDM in Data science & analytics is very good and I think it's a very solid course. I learned a lot.



TRIPTI SINGH
Software Engineer



CHANDAN KUMAR JHA
Data Consultant

“Not only did the course (PGDM in Data science & analytics) teach me the proper techniques of Data analytics and the process of data handling as promised, but it also taught me how to learn independently, how to stick with a problem and find ways of solving it, and perhaps most significantly, the experience taught me the skills that will enable me to continue to develop my profile beyond the electronic classroom and helped a lot to keep up my skills as per industry standards.



Detailed curriculum

Data Modelling Concepts

PC1: Data modelling should adhere to principles of clarity, consistency, accuracy, completeness, and reusability.

PC2: Data modelling includes improved data quality, understanding, access, and decision-making.

DBMS and RDBMS (MY SQL)

PC3: DBMS and RDBMS should effectively manage, organize, and retrieve data while ensuring data integrity and security.

PC4: DBMS and RDBMS should provide efficient data access, query processing, and transaction management capabilities. Hadoop Installation

RDBMS Functionality

PC5: RDBMS should effectively organize and store data in tables, optimize data retrieval and analysis, and maintain data integrity and consistency.

PC6: RDBMS should facilitate data segregation, joining, and aggregation to enhance query performance and minimize redundancy

NoSQL (MongoDB)S

PC7: Efficiently store and retrieve data with high scalability and availability while providing flexibility in data modelling.

PC8: High availability, replication, and consistency to ensure data integrity and reliability for mission-critical applications.

Introduction to Python

PC9: Learners should gain a basic understanding of Python programming concepts, including environment setup, syntax, variables, data types, operators, data structures, control flow statements, and functions.

PC10: Learners should be able to write simple Python programs to solve basic programming problems.

Functions

PC11: Functions encompass designing efficient and reusable user-defined functions, utilizing inbuilt functions effectively, and understanding the role of lambda functions in concise and functional programming.

PC12: Python date and time handling include accurately managing and formatting dates and times, working with various time zones and date-related calculations. For modules and packages, it involves organizing and importing modules and packages efficiently, maintaining clear code structure, and managing dependencies.

Classes And Objects

PC13: Working with classes and objects in Python include a thorough understanding of the concepts, the ability to create and manipulate NumPy ndarrays, perform array creation and operations, iterate over arrays effectively, and utilize indexing and slicing techniques to access data efficiently.

PC14: NumPy functions involve proficiency in using various NumPy functions and methods for data manipulation. For Pandas, performance criteria include the ability to work with Series and DataFrames, perform descriptive statistics, read and write CSV files, and efficiently index and slice DataFrames.

DataFrame Basic Functions

PC15: DataFrame Join() and GroupBy() should effectively combine and manipulate data while preserving its integrity and structure.

PC16: DataFrame Join() should efficiently combine data from multiple DataFrames based on common attributes, while GroupBy() should effectively aggregate and analyze data based on specific criteria

Data Modelling Concepts

PC1: Data modelling should adhere to principles of clarity, consistency, accuracy, completeness, and reusability.

PC2: Data modelling includes improved data quality, understanding, access, and decision-making.

